ONETECH

*Accredited Testing Laboratory under the terms of ISO/IEC 17025*

# *ENVIRONMENTAL TEST REPORT*

*for*

**Product: Side Destination Indicator**

**"MODEL NAME: SDI"**

**APPLICANT**

## Woojin Industrial Systems Co., Ltd.

**613-6 Bangchuk-ri, Sari-myun, Goesan-gun, Chungchongbuk-do, Korea**

**October 31, 2012**

**Test Results**

**"Refer to the test results attached"**

**1. As a test result of sample which was submitted from the client, this test report does not guarantee the whole product quality.**

**2. This test report should not be used for public relations, advertisement, conduct propaganda and raised for a suit without a written agreement of testing institute, and should not be used for various purpose beside subscribed use.**

**3. This test report including test result conducted by using the test method which was suggested form a client.**

**(signature)**

**Goan Woo Lee / Managing Director**

***This is only valid in connection with Test Report:* R12OR-038**

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# TEST REPORT

| | | |
|---|---|---|
| **Report No** ..... : | R12OR-038 | |
| Date of issue ..... : | October 31, 2012 | |
| **Testing Laboratory** ..... : | ONETECH Corp. | AGR. No.: A12OA-168 |
| Address ..... : | 301-14 Daessangnyeong-ri, Chowol-eup, Gwangju-si, Gyeonggi-do 464-862 Korea | |
| Test location ..... : | Same as above | |
| **Applicant** ..... | Woojin Industrial Systems Co., Ltd. | |
| Address ..... : | 613-6 Bangchuk-ri, Sari-myun, Goesan-gun, Chungchongbuk-do, Korea | |
| Standard ..... : | Test specification supplied by the manufacturer, (Document No. 423-316, Section No. 7.3) | |
| Purpose of use ..... : | Quality Control | |
| Number of pages (Report)..... : | 9 Pages | |
| Number of pages (Attachments) ..... : | N/A | |

| | | | |
|---|---|---|---|
| Prepared by ..... : | | Reviewed by..... : | |
| (+ signature) | Dong Kyun Park / Engineer | (+ signature) | Jin Won Seo / Project Engineer |

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# 1. CLIENT AND MANUFACTURER INFORMATION

-. Client : Woojin Industrial Systems Co., Ltd.

-. Address : 613-6 Bangchuk-ri, Sari-myun, Goesan-gun, Chungchongbuk-do, KOREA

-. Manufacturer : Woojin Industrial Systems Co., Ltd.

-. Address : 613-6 Bangchuk-ri, Sari-myun, Goesan-gun, Chungchongbuk-do, KOREA

-. Telephone No. : +82-43-820-4263

-. Fax No. : +82-43-836-7653

# 2. TEST SUMMARY

## 2.1 Test items and results

| STANDARDS | | RESULTS |
|---|---|---|
| Test specification supplied by the manufacturer (Document No. 423-316, Section No. 7.3) | Damp Heat Test | PASS |

## 2.2 Criterion description

| Measurement | |
|---|---|
| Initial measurement | It checks normal operating of the sample before test starts. |
| Intermediate measurement | After the first cyclic is finished, it checks normal operating of the sample with enough dry status. |
| Final measurement | It checks normal operating of the sample with enough dry status at the end of test. |

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## 2.3 Basis test condition

| Items | | Descriptions |
|---|---|---|
| Initial Measurements | | - No crack, transformation and swell should be visually not found.<br>- After the EUT is connected to peripheral equipments stated in the section 3.2, the EUT should be no differences in functions of operation stated in the section 3.3. |
| Precondi-tioning | Regulations | The regulations of preconditioning is not required : ■<br>The regulations of preconditioning is required : □ |
| | Contents | N/A |
| Intermediate Measurements | Regulations | The regulation of intermediate measurements is not required : □<br>The regulation of intermediate measurements is required : ■ |
| | Contents | - After the EUT is connected to peripheral equipments stated in the section 3.2, the EUT should be no differences in functions of operation stated in the section 3.3. |
| Recovery | Regulations | The regulation of recovery is not required : ■<br>The regulation of recovery is required : □ |
| | Contents | N/A |
| Final Measurements | | - No crack, transformation and swell should be visually not found.<br>- After the EUT is connected to peripheral equipments stated in the section 3.2, the EUT should be no differences in functions of operation stated in the section 3.3. |

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# 3. EUT (Equipment Under Test)

## 3.1 Identification of EUT

-. Equipment : Side Destination Indicator

-. Type/model : SDI

-. Brand Name : N/A

-. Serial number : N/A

-. Manufacturer : Woojin Industrial Systems Co., Ltd.

## 3.2 Configuration

| Model | Manufacturer | Description | Connected to |
|---|---|---|---|
| SDI | Woojin Industrial Systems Co., Ltd. | Side Destination Indicator (EUT) | - |

## 3.3 Operational mode of EUT

- After the EUT is connected to peripheral equipments stated in the section 3.2, the EUT maintains normal operations.

## 3.4 EUT modifications

- N/A

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# 4. DAMP HEAT TEST

The **Damp Heat Test** was performed at ONETECH Corp. **Date** : October 19 ~ October 21, 2012

## 4.1 Environmental condition

| | |
|---|---|
| Ambient temperature | 22 ℃ |
| Relative humidity | 50 % R.H. |
| Air pressure | 101 kPa |

## 4.2 Test equipment used

| | Model Number | Manufacturer | Description | Serial Number |
|---|---|---|---|---|
| ■ | ER-55MHHP | HITACHI | TEMPERATURE AND HUMIDITY CHAMBER | U5972178 |

## 4.3 Test condition

### 4.3.1 Detailed test condition

| Items | | Descriptions |
|---|---|---|
| Standard | | Test specification supplied by the manufacturer |
| Operational mode of EUT | | Non-operational test : ■ Operational test : □ |
| Method for change of temperature | | Rapid change of temperature : □ Regular change of temperature : ■ |
| Status of heat -dissipating | | Heat-dissipating : □ Non heat-dissipating : ■ |
| Initial Measurements | | There is same as the initial measurements stated in the section 2.3. |
| Precondi-tioning | Regulations | The regulations of preconditioning is not required : ■<br>The regulations of preconditioning is required : □ |
| | Contents | N/A |
| Intermediate Measurements | Regulations | The regulation of intermediate measurements is not required : □<br>The regulation of intermediate measurements is required : ■ |
| | Contents | See the next graph |
| Recovery | Regulations | The regulations of recovery is not required : ■<br>The regulations of recovery is required : □ |
| | Contents | N/A |
| Final Measurements | | There is same as the final measurements stated in the section 2.3. |

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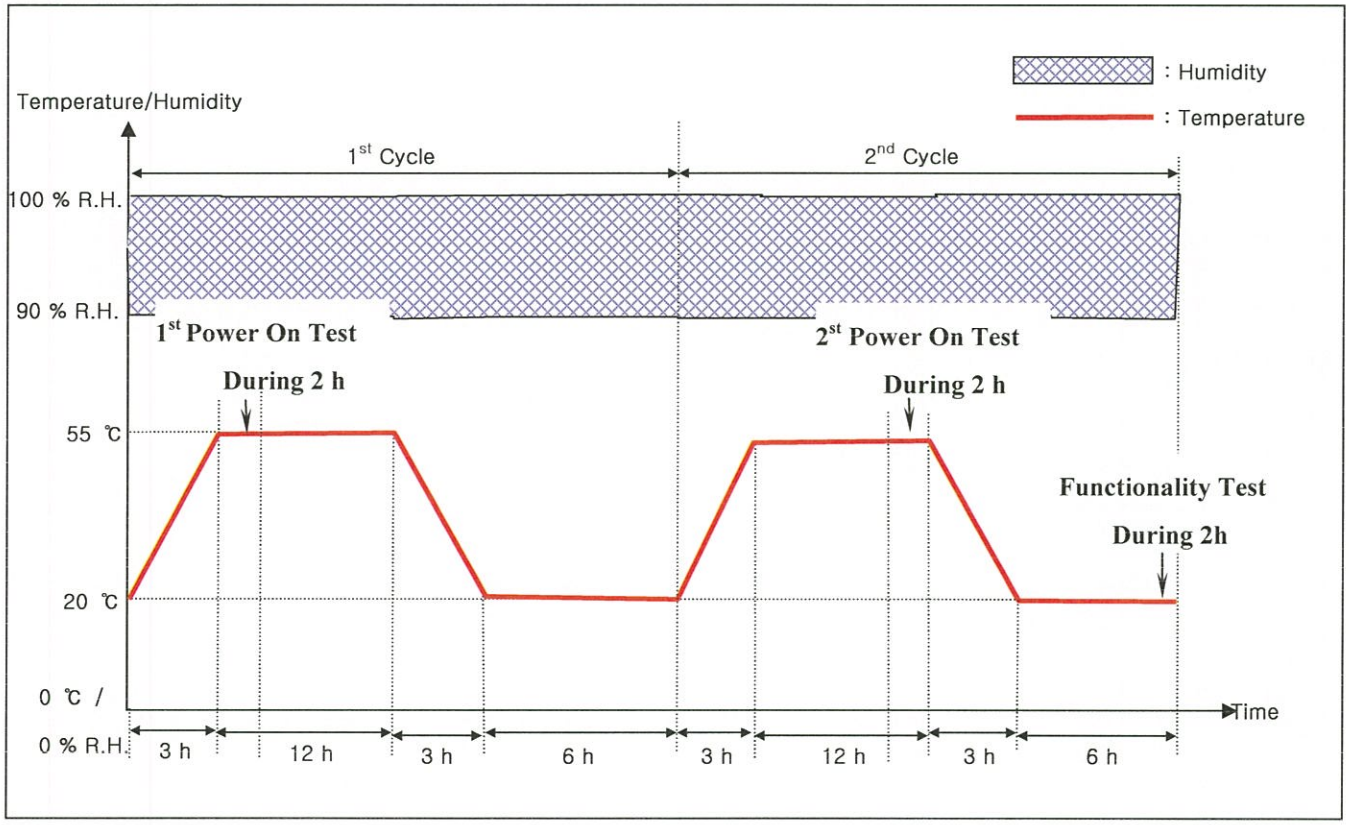
### 4.3.2 Test severities

| Temperature | | | Humidity | | | Cycle Number | Test sequence |
|---|---|---|---|---|---|---|---|
| Set-up | | Rate of change | Set-up | | Rate of change | | |
| Temp-erature | Time | | Hum-idity | Time | | | |
| 20 ℃<br>55 ℃ | 12 h<br>24 h | See the below graph | 95 % R.H. | 48 h | See the below graph | 2 | [55 ℃(95 % R.H.) → 20 ℃(95 % R.H.)] = 2 Cycle |

**<Remark : Damp Heat Test Graph>**



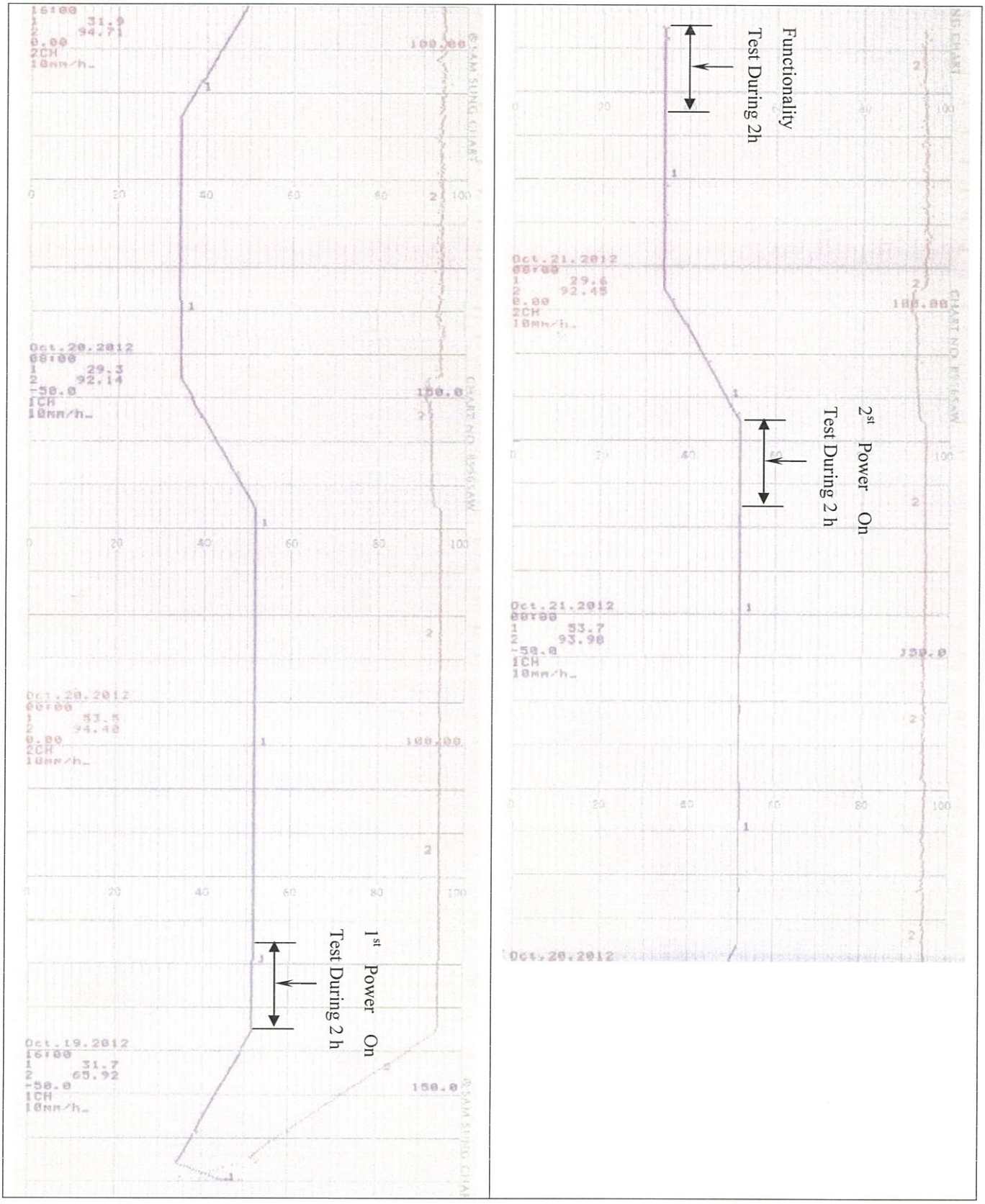

## 4.4 Test result

According to criterion description stated in the section 2.3, the initial measurements, intermediate measurements and final measurements were measured as follows.

| EUT | Initial Measurements (1st Power On Test) | Intermediate Measurements (2st Power On Test) | Final Measurements (Functionality Test) |
|---|---|---|---|
| Sample 1 | PASS | PASS | PASS |

**Tested by: Dong-Kyun Park / Engineer**

**4.5 Test data**


Functionality
Test During 2h
2st Power On
Test During 2 h
1st Power On
Test During 2 h

## APPENDIX I - PHOTO OF CHAMBER TEST

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